

ORIGINAL RESEARCH ARTICLE

Incidence and Predictors of Extracranial Bleeding on Oral Anticoagulants for Stroke Prevention in Patients With Atrial Fibrillation: A COMBINE-AF Analysis

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BACKGROUND: Extracranial bleeding is the most common complication of oral anticoagulant (OAC) therapy for atrial fibrillation (AF), but its clinical importance for patients may be underrecognized. We sought to characterize extracranial bleeding events according to standardized severity definitions, identify baseline risk factors for bleeding, and quantify their population attributable fraction in patients with AF receiving OACs.

METHODS: We analyzed patients receiving OACs from 5 pivotal randomized trials testing a direct OAC or warfarin in patients with AF (COMBINE-AF [A Collaboration Between Multiple Institutions to Better Investigate Non-Vitamin K Antagonist Oral Anticoagulant Use in Atrial Fibrillation]). The primary outcome was extracranial clinically relevant bleeding, defined as a first episode of extracranial major or clinically relevant nonmajor bleeding according to International Society on Thrombosis and Haemostasis criteria. The Kaplan-Meier method was used to calculate the cumulative incidence of bleeding by category. Multivariable Cox regression models were used to estimate adjusted hazard ratios (HRs) with 95% CI. Logistic regression models were used to calculate average population attributable fraction with 95% CI.

RESULTS: Of 73 737 patients treated with OACs, 10 634 experienced clinically relevant extracranial bleeding over a mean follow-up of 705 days (cumulative incidence, 26% [95% CI, 18%–35%]; 7.6 per 100 person-years). This included 3188 major bleeds (cumulative incidence, 7% [95% CI, 6%–7%]; 2.1 per 100 person-years) and 7446 clinically relevant nonmajor bleeds (cumulative incidence, 19% [95% CI, 12%–28%]; 5.2 per 100 person-years). The distribution of bleeding sites differed by severity, with gastrointestinal bleeds comprising 26% of clinically relevant bleeds, 49% of major bleeds, and 15% of clinically relevant nonmajor bleeds. Risk factors for extracranial bleeding were consistent across severity bleeding categories, and baseline covariates in our multivariable models accounted for 66% to 69% of the population attributable bleeding risk.

CONCLUSIONS: Extracranial clinically relevant bleeding is common among patients with AF treated with OACs and may more accurately reflect the overall burden of bleeding than major bleeding alone. Our models explained about two-thirds of the average population attributable risk, suggesting that additional unmeasured or unknown factors contribute to bleeding risk.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: atrial fibrillation ■ bleeding risk factors ■ clinically relevant bleeding ■ COMBINE-AF ■ direct oral anticoagulants ■ extracranial bleeding ■ individual patient meta-analysis ■ major bleeding ■ oral anticoagulation/ anticoagulant therapy ■ population attributable fraction ■ warfarin

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Clinical Perspective

What Is New?

- In anticoagulated patients with atrial fibrillation, extracranial clinically relevant bleeding affects 1 in 4 patients over 2 years of follow-up and may better capture the broader burden of bleeding than major bleeding alone.
- Despite differences in severity, major and clinically relevant nonmajor bleeding have broadly similar risk factors, suggesting that they may arise from common pathophysiological mechanisms.
- Baseline clinical factors accounted for only about two-thirds of the average population attributable risk of bleeding, suggesting that important determinants of bleeding are not captured by routinely collected baseline characteristics.

What Are the Clinical Implications?

- Knowledge of bleeding risk factors and the broader burden of bleeding can help clinicians better counsel patients about anticoagulation therapy.
- Clinically relevant nonmajor bleeding should not be overlooked because it contributes importantly to the overall burden of bleeding.
- Addressing potentially reversible contributors, such as anemia, concomitant antiplatelet or nonsteroidal anti-inflammatory drug use, and anticoagulant choice, may help reduce the risk of bleeding.

Nonstandard Abbreviations and Acronyms

aPAF	average population attributable fraction
AF	atrial fibrillation
DOAC	direct oral anticoagulant
HR	hazard ratio
ISTH	International Society on Thrombosis and Haemostasis
OAC	oral anticoagulant
TIA	transient ischemic attack

Oral anticoagulant (OAC) therapy is highly effective at reducing thromboembolic complications in patients with atrial fibrillation (AF), but bleeding remains the main complication, limiting its use. OAC-related bleeding is the most common adverse drug event that leads to emergency department visits, hospitalization, and death.^{1–3}

Accurate assessment of potential benefits and harms about OAC therapy is central to informed, evidence-based decision-making. Although uncommon, intracranial bleeding may be overemphasized when considering the potential harms, given substantial short-term mortality and permanent functional impairment.⁴ Extracranial

bleeds are much more common but less well characterized, and their importance is often underappreciated despite mortality rates of up to 10% to 15% reported within 30 days of bleeding.^{5–7}

Commonly used definitions for major bleeding (eg, International Society on Thrombosis and Haemostasis (ISTH)) are frequently reported in clinical trials but do not fully capture the impact of bleeding on patients.⁸ In addition to the 2% to 4% of patients who experience major bleeding on OAC therapy annually, another 4% to 12% of patients develop less severe clinically relevant nonmajor (CRNM) bleeding, which may impact quality of life, treatment adherence, and health care use.^{6,9–13} Clinically relevant bleeding, which includes both major and CRNM bleeding, better captures the breadth of patient-important bleeding events and is increasingly adopted as the primary safety outcome in antithrombosis trials.^{14–17} However, the use of more inclusive bleeding definitions may have implications for the interpretation of results of randomized trials and their application in clinical practice. A better understanding of the nature and importance for patients of clinically relevant bleeding is needed.

The COMBINE-AF database (A Collaboration Between Multiple Institutions to Better Investigate Non-Vitamin K Antagonist Oral Anticoagulant Use in Atrial Fibrillation) includes individual patient data from pivotal randomized trials comparing direct OACs (DOACs) with warfarin or aspirin in AF.¹⁸ Using this dataset, we sought to characterize extracranial bleeding events according to definitions of clinically relevant, major, and CRNM bleeding; identify baseline risk factors; and determine their average population attributable fraction (aPAF) in patients with AF receiving OACs.

METHODS

Anonymized data from COMBINE-AF are shared by COMBINE-AF executive committee members and their corresponding institutions. These data cannot be shared outside of these institutions due to preexisting data privacy restrictions and contractual agreements. Individual researchers may contact the COMBINE-AF executive committee to discuss opportunities for collaboration.

Design

The design and rationale of COMBINE-AF have been described previously.¹⁸ In brief, COMBINE-AF includes individual patient data from 77 282 deidentified patients from 5 pivotal randomized clinical trials that compared DOACs with warfarin or aspirin in patients with AF. For this analysis, we included all patients receiving OACs from the RE-LY (Randomized Evaluation of Long-term Anticoagulation Therapy), ROCKET AF (Rivaroxaban Once Daily Oral Direct Factor Xa Inhibition Compared with Vitamin K Antagonism for Prevention of Stroke and Embolism Trial in Atrial Fibrillation), ARISTOTLE (Apixaban for Reduction in Stroke and Other Thromboembolic Events in Atrial Fibrillation), ENGAGE AF-TIMI 48 (Effective

Anticoagulation With Factor Xa Next Generation in Atrial Fibrillation–Thrombolysis in Myocardial Infarction 48), and AVERROES (A Phase III Study of Apixaban in Patients With Atrial Fibrillation) trials. Institutional review board approval has been obtained for all COMBINE-AF component trials, and the subjects provided written informed consent.

Study Population

The on-treatment population was analyzed, as defined by each individual trial, typically including participants who received at least 1 dose of the study drug and were followed for events occurring from the start of treatment up to 2 days after permanent discontinuation.

Outcomes

Outcome definitions in COMBINE-AF have been described previously.¹⁸ The primary outcome of this analysis was extracranial clinically relevant bleeding, defined as a first episode of extracranial major or CRNM bleeding. As shown in Table S1, bleeding outcomes in individual trials were generally defined as major bleeding and CRNM bleeding using criteria defined by the ISTH.^{8,13} For the RE-LY trial, ISTH-defined CRNM bleeding was not collected, so we included extracranial bleeding that was associated with hospitalization or transfusion. In all of the trials, bleeding events were adjudicated independently by assessors blinded to treatment assignment. The secondary outcomes were (1) extracranial major bleeding and (2) extracranial CRNM bleeding.

Statistical Analyses

Baseline characteristics were summarized using descriptive statistics with mean and SD or median and interquartile range for continuous variables and proportions for categorical variables. Crude event rates of bleeding complications were reported overall and separately according to bleeding category and primary bleeding site. Bleeding characteristics were reported descriptively. We calculated the cumulative incidence of extracranial clinically relevant bleeding, major bleeding, and CRNM bleeding while accounting for the competing risk of death.

We examined risk factors separately for extracranial clinically relevant, major, and CRNM bleeding. Candidate predictors were selected on the basis of clinical relevance to identify factors associated with each type of bleeding. Hemoglobin was modeled as a categorical variable on the basis of analysis using restricted cubic splines, which showed an inflection point at approximately 12 g/dL, below which the risk of bleeding increased. Variables in the final multivariable model were selected using bidirectional stepwise selection, which combines forward and backward selection; variables were iteratively added to the model on the basis of statistical significance and subsequently evaluated for removal if they no longer met the prespecified significance threshold. This process continued until all variables retained in the model met the specified inclusion criterion ($P < 0.10$). We also conducted a sensitivity analysis using least absolute shrinkage and selection operator–based variable selection.

We conducted a sensitivity analysis, excluding the RE-LY trial, which did not collect CRNM bleeding. Variables with

missingness of $>5\%$ of total sample size were not included in the model. Several variables had substantial missingness because they were not collected in some of the included trials (Hispanic ethnicity, history of liver disease, history of coronary artery bypass grafting, history of percutaneous intervention, prior hemorrhagic stroke, history of gastrointestinal bleeding, and history of nongastrointestinal bleeding), resulting in missingness at the trial level rather than sporadic missing observations. This cutoff was selected because this level of missingness $<5\%$ of total sample size is generally considered unlikely to introduce meaningful bias or significantly affect model estimates in regression analyses.^{19–21} We conducted a sensitivity analysis using the multiple imputation by chained equations approach. To assess for multicollinearity, we calculated correlation coefficients between variables in each of the final models and then estimated variance inflation values. Hazard ratios (HRs) with 95% CI were reported.

We calculated aPAF and 95% CI of risk factors by fitting a logistic model using the methods of Ferguson and O'Connell.²² We chose the Ferguson method because it permits multivariable average PAF estimation, thereby estimating the contribution of each risk factor individually and collectively, accounting for overlap between risk factors. While extensions of PAF methodology to time-to-event settings can accommodate multiple exposures, they are more complex and less routinely implemented. Our use of logistic regression was therefore driven by the goal of estimating a combined/average PAF across overlapping risk factors in a more practical and interpretable way. When calculating aPAF, we adjusted for the same covariates as the multivariable Cox model. As a sensitivity analysis, we also explored Cox-based PAF methods in addition to our primary logistic regression–based PAF analysis.

Statistical analyses were conducted in Statistical Analysis System version 9.4.

RESULTS

Extracranial Clinically Relevant Bleeding (Major Bleeding or CRNM Bleeding)

Baseline characteristics for patients with and without extracranial clinically relevant bleeding are depicted in Table 1. Table S2 shows the baseline characteristics for patients with extracranial major or CRNM bleeding. Of 73 737 patients, 10 634 patients (cumulative incidence, 26% [95% CI, 18%–35%]; incidence rate, 7.6 per 100 person-years; Figure 1; Table 2) experienced extracranial clinically relevant bleeding. Average follow-up was 705 days (SD, 317). Patients were randomized to standard-dose DOAC ($n=4344$, 40.9%), lower-dose DOAC ($n=1816$, 17.1%), or warfarin ($n=4474$, 42.1%). Compared with patients with no bleeding, those who experienced extracranial clinically relevant bleeding were generally older; had more medical comorbidities, including coronary artery disease, diabetes, cancer, and prior bleeding; and had a higher frequency of phytonadione antagonist use before enrollment in the individual trials.

The most frequent extracranial clinically relevant bleeding events were gastrointestinal (26%), skin/

Table 1. Baseline Characteristics of Patients With and Without Bleeding

Characteristics	Clinically relevant bleeding (n=10 634)	No bleeding (n=54 664)	Standardized difference*
Age, y, median (IQR)	74.0 (67.0,79.0)	71.0 (64.0,77.0)	0.29105
Female sex, n (%)	3825 (36.0)	20 552 (37.6)	0.03513
Race, n (%)			...
White	8502 (80.0)	44 103 (80.7)	0.00836
Black	133 (1.3)	649 (1.2)	0.00934
Asian	1554 (14.6)	7761 (14.2)	0.00937
Other	445 (4.2)	2147 (3.9)	0.00484
Hispanic	807 (11.1)	5930 (14.2)	0.07216
Physical measures			
Body weight, kg, median (IQR)	81.0 (70.0, 94.7)	81.0 (70.0, 94.1)	0.01622
Body weight category, kg, n (%)			...
<60	988(9.3)	5263(9.6)	0.01018
60–, <120	9107 (85.7)	46 769 (85.7)	0.00112
≥120	534 (5.0)	2569 (4.7)	0.01198
Systolic BP, mm Hg, median (IQR)	130.0 (120.0, 140.0)	130.0 (120.0, 140.0)	0.06240
Body mass index, kg/m ² , median (IQR)	28.4 (25.2, 32.4)	28.3 (25.2, 32.1)	0.03137
Body mass index, kg/m ² , n (%)			...
<25	2511 (23.7)	12 982 (23.8)	0.00531
25–, <30	3938 (37.1)	20 880 (38.3)	0.02552
≥30	4154 (39.2)	20 650 (37.9)	0.03007
Randomized treatment, n (%)			
Warfarin	4474 (42.1)	20 813 (38.1)	0.07108
Standard-dose DOAC	4344 (40.9)	24 286 (44.4)	0.05858
Lower-dose DOAC	1816 (17.1)	9565 (17.5)	0.01563
Medical history, n (%)			
Chronic liver disease	496 (5.6)	1641 (4.1)	0.06535
Cancer	1049 (9.9)	3371 (6.2)	0.10929
Falls	715 (6.8)	2477 (4.7)	0.06646
Diabetes	3801 (35.7)	16 203 (29.6)	0.13434
Hypertension	9524 (89.6)	47 974 (87.8)	0.07081
Coronary artery disease	3705 (35.3)	15 053 (28.9)	0.13304
Prior myocardial infarction	1768 (16.8)	7267 (13.9)	0.07048
CABG	997 (11.2)	2290 (5.7)	0.18454
PCI	968 (10.8)	2875 (7.2)	0.11653
Valvular heart disease	2249 (21.1)	10 064(18.4)	0.05745
Heart failure	5255 (49.4)	25731 (47.1)	0.07423
Stroke, any	2174 (20.4)	9650 (17.7)	0.07291
TIA	1417 (13.3)	6131 (11.2)	0.06118
Stroke and/or TIA	3299 (31.0)	14 699 (26.9)	0.09215
Stroke, hemorrhagic	8 (0.1)	38 (0.2)	0.01221
Paroxysmal AF	2613 (24.6)	12 258 (22.4)	0.03423
Smoking, ever	5052 (47.5)	22 746 (41.6)	0.09435
Prior GI bleeding	575 (6.4)	1215 (3.0)	0.14985
Prior non-GI bleeding	893 (10.0)	2597 (6.5)	0.11162
F-18 frailty score, median (IQR)†	6.0 (5.0,7.0)	5.0 (4.0,6.0)	0.31800

(Continued)

Table 1. Continued

Characteristics	Clinically relevant bleeding (n=10 634)	No bleeding (n=54 664)	Standardized difference*
CHADS ₂ score, median (IQR)	3.0 (2.0,4.0)	2.0 (2.0,3.0)	0.30056
CHADS ₂ score, n (%)			...
0–1	798 (7.5)	10 219 (18.7)	0.34836
2	3645 (34.3)	19 069 (34.9)	0.00863
3 or greater	6191 (58.2)	25 374 (46.4)	0.24359
CHADSVasc score, median (IQR)	4.0 (3.0, 5.0)	4.0 (3.0, 5.0)	0.34157
CHADSVasc score, n (%)			...
0–2	922 (8.7)	10 521 (19.2)	0.30632
3–5	7491 (70.4)	36 152 (66.1)	0.08691
6 or greater	2221 (20.9)	7986 (14.6)	0.16951
DOAC score, median (IQR)†	6.0 (5.0, 8.0)	5.0 (4.0, 7.0)	0.42256
HASBLED score, median (IQR)	3.0 (2.0, 3.0)	2.0 (2.0, 3.0)	0.32673
Baseline Medication, n (%)			
Prior phytonadione antagonist use, ever	7739 (72.8)	35 798 (65.5)	0.14266
Aspirin/clopidogrel	4225 (39.7)	18 203 (33.3)	0.11492
Thienopyridine	404 (3.8)	1390 (2.5)	0.05715
β-Blocker	6810 (64.2)	35 027 (64.4)	0.00250
Calcium channel blocker	3458 (32.6)	16 156 (29.7)	0.05225
NSAID	558 (5.2)	2153 (3.9)	0.03928
Cox II inhibitor	72 (0.7)	260 (0.5)	0.01390
Digoxin	3266 (30.8)	17 699 (32.6)	0.02887
PPI	1661 (15.8)	5499 (10.6)	0.13467
Amiodarone	1055 (9.9)	5962 (11.0)	0.02907
Laboratory studies			
Creatinine clearance, mL/min, median (IQR)	65.0 (50.9, 84.1)	71.5 (55.0, 92.1)	0.17855
Creatinine clearance			...
<50 mL/min	2484 (23.4)	9686 (17.8)	0.13228
50–80 mL/min	5037 (47.4)	24 146 (44.3)	0.05422
>80 mL/min	3104 (29.2)	20 722 (38.0)	0.17150
Hemoglobin, g/dL, median (IQR)	13.9 (12.8, 14.9)	14.2 (13.2, 15.2)	0.22116
Platelet count×10 ⁹ /L, median (IQR)	204.0 (169.0, 245.0)	212.0 (178.0, 252.0)	0.11586

For the RE-LY trial, bleeds associated with hospitalization or red blood cell transfusion that did not qualify as major bleeds were classified as clinically relevant nonmajor bleeding. AF indicates atrial fibrillation; BP, blood pressure; CABG, coronary artery bypass grafting; CHADS₂, xxx; CHADSVasc, xxx; DOAC, direct oral anticoagulant; GI, gastrointestinal; HASBLED, xxx; IQR, interquartile range; NSAID, nonsteroidal anti-inflammatory drug; PCI, percutaneous intervention; PPI, xxx; and TIA, transient ischemic attack.

*Standardized difference applies to clinically relevant bleeding versus minor bleeding/no bleeding.

†Frailty score.³⁷

‡Direct oral anticoagulant score.³⁸

hematoma (19%), and genitourinary (14%), as shown in Table 2 and Figure 2. Table S3 shows sites of extracranial clinically relevant bleeding according to randomized treatment. Of the 10 634 patients with extracranial clinically relevant bleeding, 3187 (30%) were hospitalized, of whom 572 (5%) were admitted to the intensive care unit. One-hundred twenty-five (1%) extracranial clinically relevant bleeds were fatal. Red blood cell transfusion was administered to 15% of patients, and surgery/invasive procedures were conducted in 20%.

Risk Factors and PAF

Multivariable Cox regression analysis is shown in Table 3 and Figure 3. In the adjusted analysis, the predictors of clinically relevant extracranial bleeding with the largest magnitude of effect were hemoglobin <12 g/dL (HR, 1.48 [95% CI, 1.39–1.58]) and creatinine clearance <60 mL/min (HR, 1.30 [95% CI, 1.21–1.41]) at baseline, Asian ancestry (HR, 1.30 [95% CI, 1.22–1.39]), prior stroke/transient ischemic attack (TIA) (HR, 1.28 [95%

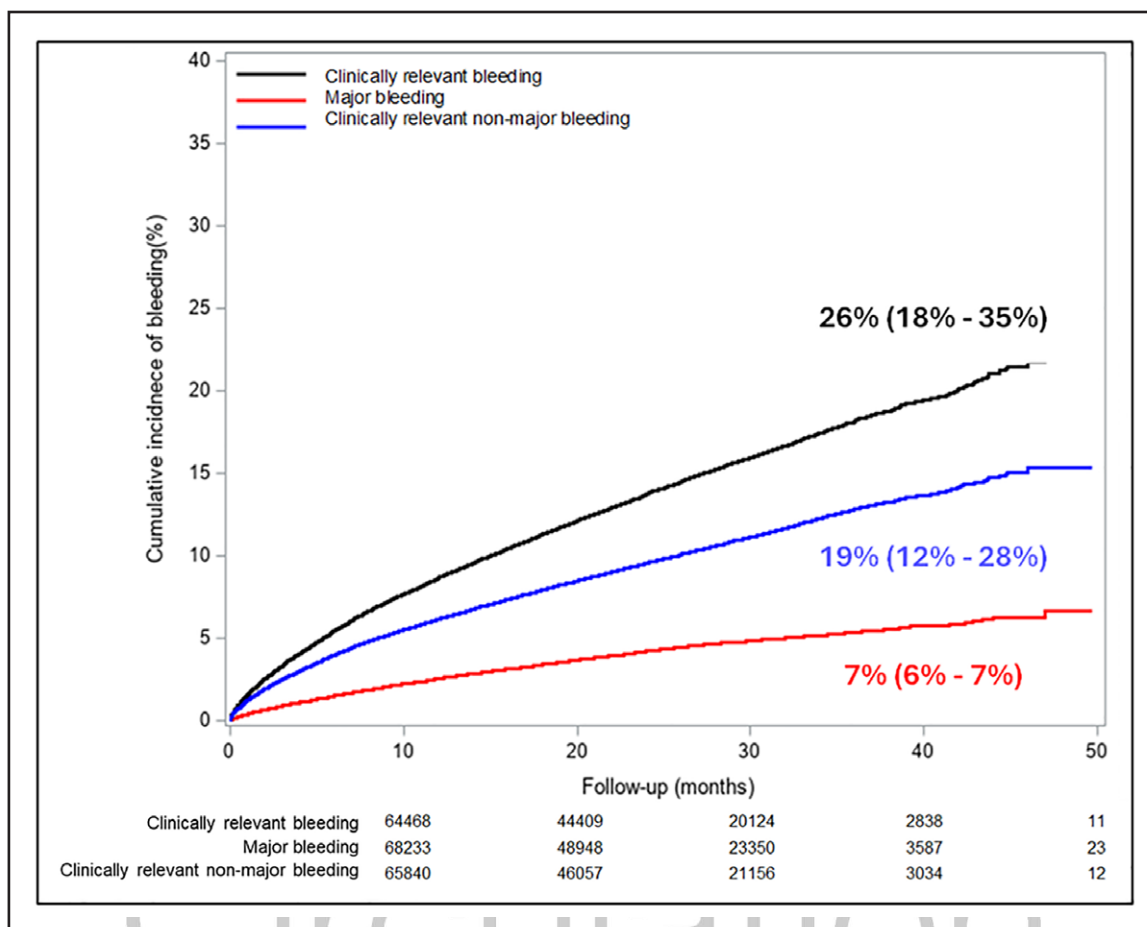


Figure 1. Cumulative incidence of bleeding according to bleeding category.

CI, 1.22–1.33], and history of cancer (HR, 1.29 [95% CI, 1.20–1.38]). Both standard-dose (HR, 0.91 [95% CI, 0.86–0.95]) and lower-dose DOAC (HR, 0.75 [95% CI, 0.71–0.79]) were associated with a decreased risk of extracranial clinically relevant bleeding compared with warfarin, as was female sex (HR, 0.95 [95% CI, 0.89–1.00]). Variance inflation values were <3.5, suggesting no multicollinearity. The proportional hazard assumption was not violated. A sensitivity analysis using the multiple imputation by chained equations approach to missing observations showed results consistent with those of the primary analysis, with similar effect estimates and the same overall pattern of significant predictors (Table S4). We also performed least absolute shrinkage and selection operator–based variable selection as a sensitivity analysis. The results were highly consistent with those obtained using the stepwise approach. The same variables were selected in both models, except for calcium channel blocker use, which was also selected by the least absolute shrinkage and selection operator procedure. The overall pattern of associations remained unchanged.

Hypertension had the highest aPAF (7.47% [95% CI, 4.52%–10.42%]), followed by randomized treat-

ment (6.23% [95% CI, 3.66%–8.80%]), lower platelet count (6.21% [95% CI, 4.56%–7.86%]), smoking (ever, 6.02% [95% CI, 4.78%–7.27%]), and higher body weight (5.44% [95% CI, 4.14%–6.73%]). The overall model aPAF was 65.92% (95% CI, 61.73%–70.10%). In a sensitivity analysis, individual PAF estimates using a Cox model–based approach were broadly similar for most variables. The Cox model produced higher PAF estimates for randomized treatment (Table S5).

Extracranial Major Bleeding

Major extracranial bleeding occurred in 3188 (4%) patients with a cumulative incidence of 7% (95% CI, 6%–7%) and incidence rate of 2.1 per 100 person-years. The most common major bleeding site was the gastrointestinal tract (48%) (Table 2; Figure 2). Major bleeding was associated with hospitalization in 3187 (53%) patients, of whom 404 (13%) were admitted to the intensive care unit. Fatal bleeding occurred in 125 patients (4%) with major bleeding. Red blood cell transfusion was administered to 1498 (47%) patients, and surgery/procedures were performed in 1039 (33%).

Table 2. Characteristics of Bleeding Events

Outcome		Clinically relevant bleeding (n=10 634)	Major bleeding (n=3188)	Clinically relevant nonmajor bleeding (n=7446)
Rates of bleeding				
Crude event rate, n (%)		10 634/73 737 (14.4)	3188/73 737 (4.3)	7446/73 737 (10.1)
Incidence rate, per 100-py		10 634/139 842 (7.6)	3188/150 648 (2.1)	7446/143 548 (5.2)
Cumulative incidence, % (95% CI)		26.2 (18–35)	6.5 (6–7)	19.5 (12–28)
Bleeding site				
Chest, abdomen, pelvis	Abdominal/pelvic/retroperitoneal	65 (0.7)	65 (2.1)	...
	Cardiac	15 (0.2)	15 (0.5)	...
	Thoracic	46 (0.5)	37 (1.2)	9 (0.1)
Surgery, trauma	Surgery	436 (4.4)	266 (8.5)	170 (2.5)
	Trauma	2 (0.0)	1 (0.0)	1 (0.0)
Head and neck, respiratory	ENT	1380 (13.9)	99 (3.2)	1281 (18.8)
	Oral	362 (3.6)	7 (0.2)	355 (5.2)
	Respiratory	295 (3.0)	22 (0.7)	273 (4.0)
Skin/hematoma		2026 (20.4)	222 (7.2)	1804 (26.5)
Gastrointestinal		2621 (26.4)	1544 (49.5)	1077 (15.8)
Genitourinary		1396 (14.1)	164 (5.3)	1232 (18.1)
MSK including compartment syndrome		184 (1.9)	135 (4.3)	49 (0.7)
Other	Ophthalmological	526 (5.3)	186 (6.0)	340 (5.0)
	Other	349 (3.5)	147 (4.7)	202 (3.0)
	Spinal	11 (0.1)	11 (0.4)	...
	Vascular	11 (0.1)	8 (0.3)	3 (0.0)
Multiple		76 (0.8)	61 (2.0)	15 (0.2)
Missing/unknown	Missing	702	67	635
	Unknown	131 (1.3)	131 (4.2)	...
Bleed characteristics, n (%)				
Fatal bleeding		125 (1.2)	125 (3.9)	0 (0.0)
Transfusion of pRBCs		1591 (15.0)	1498 (47.0)	93 (1.3)
Hospitalization for bleeding*		3187 (30.0)	1681 (52.7)	1506 (20.2)
ED visit for bleeding*		1591 (15.0)	867 (27.2)	724 (9.7)
Surgery/procedure*		2076 (19.5)	1039 (32.6)	1037 (14.0)
ICU admission		572 (5.4)	404 (12.7)	168 (2.3)

ED indicates emergency department; ENT, ear, nose, throat; ICU, intensive care unit; MSK, xxx; pRBC, packed red blood cell; and py, xxx.

*Within 24 hours before and 7 days after bleed.

Risk Factors and PAF

In the adjusted multivariable analysis (Table 5 and Figure 3), the predictors with the highest magnitude of association with extracranial major bleeding were hemoglobin <12 g/dL (HR, 1.96 [95% CI, 1.77–2.18]), creatinine clearance <60 mL/min (HR, 1.78 [95% CI, 1.53–2.07]), falls (HR, 1.47 [95% CI, 1.29–1.67]), concomitant use of antiplatelet agents (HR, 1.45 [95% CI, 1.35–1.56]) and nonsteroidal anti-inflammatory drugs/cyclooxygenase II inhibitors (HR, 1.43 [95% CI, 1.35–1.56]), and smoking (ever; HR, 1.39 [95% CI, 1.28–1.50]). Lower-dose DOAC (HR, 0.82 [95% CI, 0.74–0.91]), amiodarone use (HR, 0.86 [95% CI, 0.76–0.98]) and female sex (HR, 0.92

[95% CI, 0.84–1.00]) were associated with a decreased risk of major bleeding. A sensitivity analysis using a competing risk regression approach (Fine-Gray model) in which CRNM bleeding was treated as a competing event for major bleeding showed that most variables show a very similar magnitude and direction of association with very few differences (Table S6). When CRNM bleeding was treated as a competing risk, a history of coronary artery disease increased the risk of major bleeding, while hypertension decreased the risk of major bleeding. Amiodarone was significant in the original model.

Creatinine clearance <60 mL/min had the highest aPAF (13.76% [95% CI, 8.84%–18.69%]), followed by smoking (ever; 9.19% [95% CI, 7.30%–11.09%]),

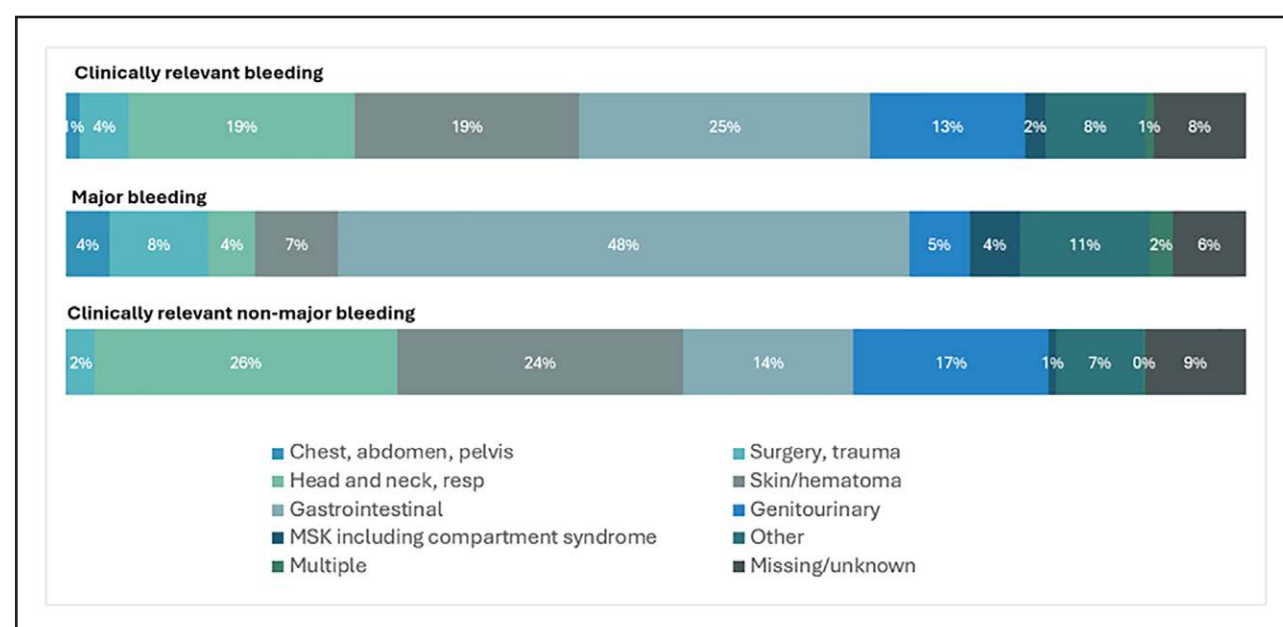


Figure 2. Distribution of bleeding sites by category of bleeding.

MSK indicates xxx.

antiplatelet use (7.84% [95% CI, 5.95–9.73]), increasing weight (7.64% [95% CI, 5.19–10.09]), and aspirin/clopidogrel use (7.43% [95% CI, 5.39–9.47]). The overall model aPAF was 69.08% (95% CI, 61.96%–76.20%). In a sensitivity analysis, individual PAF estimates using a Cox model-based approach were broadly similar for most variables. The Cox model produced higher PAF estimates for randomized treatment, weight, antiplatelet therapy, and creatinine clearance (Table S7).

Extracranial CRNM Bleeding

Extracranial CRNM bleeding occurred in 7466 (10%) patients with a cumulative incidence of 19% (95% CI, 12%–28%) and incidence rate of 5.2 per 100 person-years. The most common site of CRNM bleeding was skin/hematoma (24%) (Table 2; Figure 2). CRNM bleeding was associated with hospitalization in 1506 (20%) patients, of whom 168 (2%) were admitted to the intensive care unit. Red blood cell transfusion was administered to 93 patients (1.3%), and surgery/procedures were performed in 14%.

Risk Factors and Population Attributable Risk

Multivariable Cox regression analysis is shown in Table 5 and Figure 4. In the adjusted analysis, the predictors of extracranial CRNM bleeding with the largest magnitude of effect included Asian race (HR, 1.38 [95% CI, 1.28–1.49]), history of stroke/TIA (HR, 1.39 [95% CI, 1.32–1.46]), and cancer (HR, 1.32 [95% CI, 1.22–1.43]). Compared with warfarin, treatment with lower-dose DOAC (HR, 0.72 [95% CI, 0.67–0.77]) and

standard-dose DOAC (HR, 0.96 [95% CI, 0.85–0.95]) were associated with a decreased risk of CRNM bleeding. We conducted a sensitivity analysis, excluding the RE-LY trial, whose results were broadly consistent with those of the primary analysis shown in Table S8. The direction and magnitude of associations for most variables remained similar, except for history of hypertension and coronary artery disease, which were no longer statistically significant after exclusion of RE-LY. The overall pattern of risk factors associated with bleeding was unchanged.

Hypertension had the highest aPAF (11.67% [95% CI, 8.29%–15.05%]), followed by platelet count decrease (9.63% [95% CI, 7.40%–11.86%]), randomized treatment (7.54% [95% CI, 4.63%–10.45%]). The overall model aPAF was 69.92% (95% CI, 65.91–73.93). In a sensitivity analysis, individual PAF estimates using a Cox model-based approach were broadly similar for most variables. The Cox model produced higher PAF estimates for randomized treatment (Table S9).

DISCUSSION

In this large individual-participant meta-analysis of randomized trials, we showed that OAC-treated AF patients experience a substantial burden of extracranial clinically relevant bleeding, with a cumulative incidence of 26% over approximately 2 years of follow-up. The distribution of bleeding sites varied across bleeding categories; gastrointestinal bleeds represented 26% of clinically relevant bleeds, 49% of major bleeds, and 15% of CRNM bleeding. Overall, the clinical factors associated with highest magnitude of effect on extracranial clinically relevant

Table 3. Multivariable Cox Regression Analysis and Population Attributable Fraction: Clinically Relevant Bleeding Versus No Bleeding

Characteristics	P value	HR (95% CI)	aPAF (95% CI)*
Age, y (per 1-y increment)	<0.0001	1.03 (1.03, 1.04)	...
Age, y (per 5-y increment)	...	1.17 (1.15, 1.19)	...
Sex, female vs male	0.0597	0.95 (0.89, 1.00)	...
Race, n (%)			
White		...	...
Black	0.7817	0.97 (0.81, 1.17)	...
Asian	<0.0001	1.30 (1.22, 1.39)	...
Other	0.1424	1.08 (0.98, 1.19)	...
Height, cm	0.0058	1.00 (0.99, 1.00)	1.93 (0.33, 3.54)
Height, cm (per 5-cm increment)	...	0.98 (0.97, 0.99)	...
Weight, kg	<0.0001	1.01 (1.01, 1.01)	5.44 (4.14, 6.73)
Weight, kg (per 10-kg increment)	...	1.09 (1.07, 1.10)	...
Randomized treatment	...	...	6.23 (3.66, 8.80)
Warfarin	...	...	...
Standard-dose DOAC	<0.0001	0.91 (0.87, 0.95)	...
Lower-dose DOAC	<0.0001	0.75 (0.71, 0.79)	...
Medical history (yes vs no)			
Cancer	<0.0001	1.29 (1.20, 1.38)	1.45 (1.10, 1.89)
Falls	<0.0001	1.20 (1.11, 1.30)	0.70 (0.34, 1.06)
Diabetes	<0.0001	1.23 (1.18, 1.28)	4.72 (3.92, 5.52)
Hypertension	0.0043	1.10 (1.03, 1.17)	7.47 (4.52, 10.42)
CAD	0.0011	1.07 (1.03, 1.12)	2.23 (1.36, 3.10)
Valvular heart disease	0.0074	1.07 (1.02, 1.12)	1.09 (0.54, 1.65)
Paroxysmal AF	<0.0001	1.13 (1.08, 1.18)	1.74 (1.00, 2.47)
CHF	<0.0001	1.17 (1.12, 1.22)	4.84 (3.82, 5.87)
Stroke/TIA	<0.0001	1.28 (1.22, 1.33)	4.01 (3.37, 4.65)
Smoking (ever)	<0.0001	1.24 (1.19, 1.30)	6.02 (4.78, 7.27)
Baseline medication			
Aspirin/clopidogrel/ thienopyridine	<0.0001	1.25 (1.20, 1.31)	4.13 (3.16, 5.10)
NSAID/Cox II inhibitor	<0.0001	1.21 (1.10, 1.32)	0.50 (0.18, 0.82)
Laboratory:			
Creatinine clearance, mL/min	...	...	4.87 (1.79, 7.96)
<60	<0.0001	1.30 (1.21, 1.41)	...
60–90	<0.0001	1.16 (1.09, 1.24)	...
>90	...	...	...
Hemoglobin, g/dL, <12 vs ≥12	<0.0001	1.48 (1.39, 1.58)	2.32 (1.91, 2.74)
Platelet count per 50×10 ⁹ /L decrease	<0.0001	1.06 (1.04, 1.07)	6.21 (4.56, 7.86)
Overall aPAF			65.92 (61.73, 70.10)

Interaction testing showed a significant interaction between age and DOAC dose ($P=0.0103$) but not creatinine clearance ($P=0.2896$). AF indicates atrial fibrillation; aPAF, adjusted population attributable fraction; CAD, coronary artery disease; CHF, xxx; DOAC, direct oral anticoagulant; HR, hazard ratio; NSAID, nonsteroidal anti-inflammatory drug; and TIA, transient ischemic attack.

*Age, sex, and race were adjusted. Adjusted population attributable fraction was calculated by fitting a logistic model.

bleeding were hemoglobin <12 g/dL, creatinine clearance <60 mL/min, Asian ancestry, prior stroke/TIA, and history of cancer. Risk factors were generally consistent across bleeding categories, with some variations in the

magnitude and precision of risk estimates. Baseline covariates in our multivariable models accounted for about two-thirds of the population attributable bleeding risk. The covariates with the highest aPAF were hypertension,

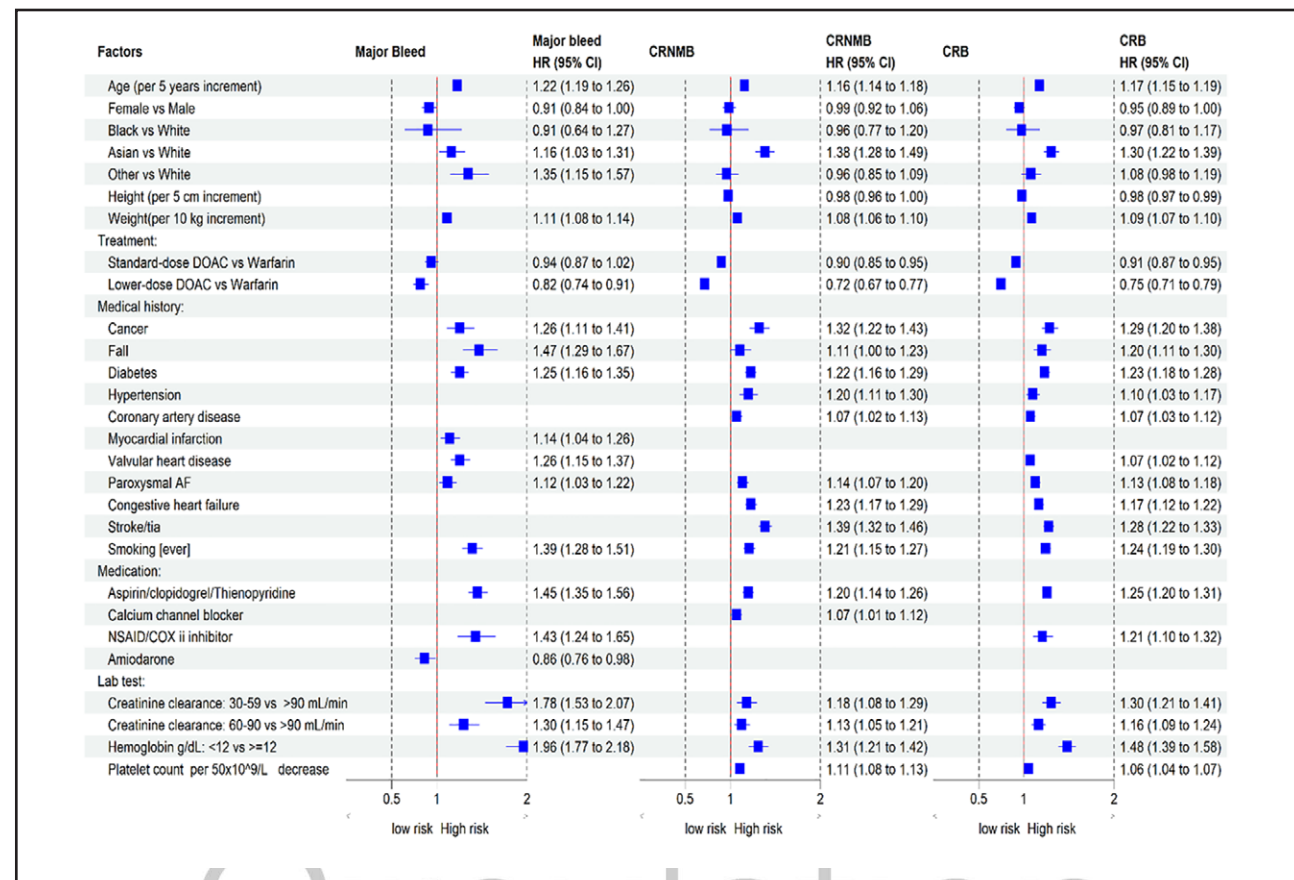


Figure 3. Multivariable Cox regression analyses of clinically relevant bleeding and major bleeding.

AF indicates atrial fibrillation; CRB, xxx; CRNMB, xxx; DOAC, direct oral anticoagulant; HR, hazard ratio; and NSAID, nonsteroidal anti-inflammatory drug.

randomized treatment, lower platelet count, smoking, and higher body weight.

We found that a first episode of extracranial CRNM bleeding occurred 2.3 times more frequently than a first episode of extracranial major bleeding, representing a substantial proportion of extracranial clinically relevant bleeding. While post-bleeding outcomes were not the focus of the present analysis, CRNM bleeding appears to worsen subsequent clinical outcomes, including overall mortality and quality of life.^{12,23} What remains uncertain is the extent to which adverse outcomes are driven by bleeding events themselves, by post-bleeding changes in OAC therapy, or by underlying factors such as older age, higher burden of multimorbidity, and frailty, which are more prevalent among patients who experience bleeding. For example, CRNM bleeding may uncover occult pathology, such as unrecognized cancer.^{24,25} Recognizing CRNM bleeding as a potential early signal of underlying pathology may therefore have important clinical implications, including prompting further diagnostic evaluation in selected patients.

While risk factors for major bleeding have been well studied, there is a paucity of data on risk factors for CRNM bleeding and, therefore, clinically relevant bleed-

ing among OAC-treated patients with AF. Major and CRNM bleeding have distinct clinical severity and prognostic implications. We showed that they have similar risk factors, suggesting that they may arise from common pathophysiologic mechanisms and potentially justifying their combination in a broader bleeding definition. Examining the downstream consequences of bleeding events, such as mortality, thromboembolism, and OAC discontinuation, is key for understanding the consequences of bleeding for patients and for justifying a broad combined bleeding outcome measure in clinical trials.

We identified risk factors for clinically relevant bleeding, such as patient characteristics (eg, sex, ancestry), physical measures (eg, increasing weight), OAC type and dose intensity, comorbidities (eg, cancer, chronic kidney disease, coronary artery disease, congestive heart failure, diabetes, falls, hypertension, smoking, stroke/TIA, valvular heart disease), concomitant medications (eg, antiplatelet therapy, nonsteroidal anti-inflammatory drugs/Cox II inhibitors), and laboratory measures (eg, hemoglobin <12 g/dL, decreasing platelet count). Although some prior studies have reported a lower risk of bleeding in overweight and obese individuals compared with individuals with normal or low body weight, other studies

Table 4. Multivariable Cox Regression Analysis and Population Attributable Fraction: Major Bleeding Versus No Bleeding

Characteristics	P value	HR (95% CI)	aPAF (95% CI)*
Age, y (per 1-y increment)	<0.0001	1.04 (1.04, 1.05)	...
Age, y (per 5-y increment)	...	1.22 (1.19, 1.26)	...
Sex (female vs male)	0.0447	0.91 (0.84, 1.00)	...
Race			
White	...	...	...
Black	0.5652	0.91 (0.64, 1.27)	...
Asian	0.0141	1.16 (1.03, 1.31)	...
Other	0.0002	1.35 (1.15, 1.57)	...
Weight, kg (per 1-kg increment)	<0.0001	1.01 (1.01, 1.01)	7.59 (5.63, 9.54)
Weight, kg (per 10-kg increment)	...	1.11 (1.08, 1.14)	...
Randomized treatment	...	...	3.10 (−1.07, 7.28)
Warfarin	...	...	...
Standard-dose DOAC	0.1385	0.94 (0.87, 1.02)	...
Lower-dose DOAC	0.0001	0.82 (0.74, 0.91)	...
Medical history (yes vs no)			
Cancer	0.0002	1.26 (1.11, 1.41)	1.33 (0.54, 2.12)
Falls	<0.0001	1.47 (1.29, 1.67)	1.62 (0.94, 2.29)
Diabetes	<0.0001	1.25 (1.16, 1.35)	5.00 (3.62, 6.38)
Myocardial infarction	0.0046	1.14 (1.04, 1.26)	1.35 (0.30, 2.41)
Vascular heart disease	<0.0001	1.26 (1.15, 1.37)	3.23 (2.13, 4.33)
Paroxysmal AF	0.0086	1.12 (1.03, 1.22)	1.61 (0.43, 2.79)
Smoking (ever)	<0.0001	1.39 (1.28, 1.51)	9.19 (7.30, 11.09)
Baseline Medication			
Aspirin/clopidogrel/ thienopyridine	<0.0001	1.45 (1.35, 1.56)	7.84 (5.95, 9.73)
NSAID/Cox II	<0.0001	1.43 (1.24, 1.65)	1.17 (0.58, 1.77)
Amiodarone	0.0268	0.86 (0.76, 0.98)	7.40 (1.22, 13.59)
Laboratory:			
Creatinine clearance	...	...	13.76 (8.84, 18.69)
<60 mL/min	<0.0001	1.78 (1.53, 2.07)	...
60–90 mL/min	<0.0001	1.30 (1.15, 1.47)	...
>90 mL/min	...	...	...
Hemoglobin, g/dL, <12 vs ≥12	<0.0001	1.96 (1.77, 2.18)	4.88 (4.01, 5.75)
Overall aPAF			69.08 (61.96, 76.20)

AF indicates atrial fibrillation; aPAF, adjusted population attributable fraction; DOAC, direct oral anticoagulant; HR, hazard ratio; and NSAID, nonsteroidal anti-inflammatory drug.

*Age, sex, and race were adjusted. Adjusted population attributable fraction was calculated by fitting a logistic model.

have shown a higher risk of major and CRNM bleeding with increasing BMI.²⁶ Several identified risk factors (eg, anemia, antiplatelet/nonsteroidal anti-inflammatory drug use, and OAC treatment choice) are actionable in clinical practice. Acknowledging the moderate predictive performance of existing clinical risk prediction scores for major bleeding, which are further limited by their overlap with stroke risk factors,²⁷ the COMBINE-AF dataset is an important potential platform for the development and validation of improved bleeding risk and net clinical benefit prediction tools in future work.

Using aPAF analysis, we estimated the relative contribution of risk factors to the overall burden of bleeding in the population. Because aPAF incorporates both the prevalence of a risk factor and the magnitude of its association with the outcome, it estimates the proportion of events attributable to that factor, thereby identifying, for the first time in this population, which risk factors contribute most to bleeding. These data directly inform bleeding risk stratification, counseling, and decision-making in the clinical setting as well as possible prevention strategies (which are currently lacking). Further, information on

Table 5. Multivariable Cox Regression Analysis and Population Attributable Fraction: Clinically Relevant Nonmajor Bleeding Versus No Bleeding

Characteristics	P value	HR (95% CI)	aPAF (95% CI)*
Age, y (per 1-y increment)	<0.0001	1.03 (1.03, 1.03)	...
Age, y (per 5-y increment)	...	1.16 (1.14, 1.18)	...
Sex, female (reference: male)	0.6998	0.99(0.92, 1.06)	...
Race			
White	...	...	...
Black	0.7390	0.96 (0.77, 1.20)	...
Asian	<0.0001	1.38 (1.28, 1.49)	...
Other	0.5439	0.96 (0.85, 1.09)	...
Height, cm	0.0131	1.00 (0.99, 1.00)	2.14 (0.25, 4.02)
Height, cm (per 5-cm increment)	...	0.98 (0.96, 1.00)	...
Weight, kg	<0.0001	1.01 (1.01, 1.01)	4.51 (3.25, 5.76)
Weight, kg (per 10-kg increment)	...	1.08 (1.06, 1.10)	...
Randomized treatment	...	...	7.54(4.63,10.45)
Warfarin	...	...	...
Standard-dose DOAC	<0.0001	0.90 (0.85, 0.95)	...
Lower-dose DOAC	<0.0001	0.72 (0.67, 0.77)	...
Medical History			
Cancer	<0.0001	1.32 (1.22, 1.43)	1.40 (0.96, 1.94)
Fall	0.0491	1.11 (1.00, 1.23)	0.34 (0.00, 0.67)
Diabetes	<0.0001	1.22 (1.16, 1.29)	4.27 (3.24, 5.30)
Hypertension	<.0001	1.20 (1.11, 1.30)	11.67 (8.29, 15.05)
CAD	0.0076	1.07 (1.02, 1.13)	2.15 (1.10, 3.19)
Paroxysmal AF	<0.0001	1.14 (1.07, 1.20)	1.78 (1.02, 2.55)
Heart failure	<0.0001	1.23 (1.17, 1.29)	6.50 (5.14, 7.85)
Stroke/TIA	<0.0001	1.39 (1.32, 1.46)	5.43 (4.64, 6.22)
Smoking (ever)	<0.0001	1.21 (1.15, 1.27)	4.85 (3.40, 6.31)
Baseline Medication			
Aspirin/clopidogrel/ thienopyridine	<0.0001	1.20 (1.14, 1.26)	2.85 (1.66, 4.04)
Calcium channel blocker	0.0127	1.07 (1.01, 1.12)	1.52 (0.55, 2.49)
Laboratory			
Creatinine clearance, mL/min	...	...	2.01 (−1.29, 5.32)
<60 mL/min	0.0004	1.18 (1.08, 1.29)	...
60–90 mL/min	0.0014	1.13 (1.05, 1.21)	...
>90 mL/min	...	...	...
Hemoglobin, g/dL, <12 vs ≥12	<0.0001	1.31 (1.21, 1.42)	1.34 (0.84, 1.84)
Platelet count (per 50×10 ⁹ /L decrease)	<0.0001	1.11 (1.08, 1.13)	9.63 (7.40, 11.86)
Overall aPAF			69.92 (65.91, 73.93)

AF indicates atrial fibrillation; aPAF, adjusted population attributable fraction; CAD, coronary artery disease; DOAC, direct oral anticoagulant; HR, hazard ratio; and TIA, transient ischemic attack.

*Age, sex, and race were adjusted. Adjusted population attributable fraction was calculated by fitting a logistic model.

the relative contribution of key clinical characteristics to bleeding risk in this population can help clinicians better contextualize bleeding risk when balancing the benefits and harms of anticoagulation, particularly in patients perceived to be at higher bleeding risk. While many identified risk factors may not be directly modifiable or challenging to modify in clinical practice, the broad range of risk fac-

tors explained about two-thirds of the population attributable bleeding risk, substantially less than similar analyses for myocardial infarction and stroke, where modifiable risk factors account for approximately 90% of the risk.^{28,29} Our findings align with a recent case-control study that showed that potentially modifiable risk factors accounted for only about one-third of the population attributable risk

for gastrointestinal bleeding in patients with cardiovascular disease.³⁰ Although our model accounted for a higher proportion of population attributable risk, a proportion of bleeding risk remains unexplained even when age, sex and race are considered. The residual unexplained bleeding risk observed in our analysis likely reflects factors that are not fully captured by traditional clinical variables (eg, frailty, vascular fragility, inflammation, occult pathology). Our findings highlight the need for novel approaches beyond standard clinical predictors to better elucidate contributors to bleeding risk to inform management. For example, the circulating biomarker GDF-15 (growth differentiation factor 15) enhances bleed risk prediction in the age–biomarkers–clinical history bleeding risk score, but GDF-15 predicts both bleeding and stroke, which may limit its use in clinical practice.³¹ Proteomic screening in patients with AF enrolled in the ARISTOTLE and RE-LY trials has identified associations between major bleeding and biomarkers of inflammation, apoptosis, oxidative stress, vascular calcification, coagulation, and fibrinolysis.³² Vascular fragility, which is associated with aging, contributes to skin bruising and bleeding and may also lead to bleeding in other vascular beds, representing a target for further investigation.^{33,34}

Our results also highlight heterogeneity in bleeding sites according to category of bleeding, underscoring the complexity of understanding the risk and consequences associated with different types of bleeding. For example, prior studies have shown that patient characteristics, bleeding time course, and post-bleed outcomes differ substantially according to bleeding site.³⁵ Resumption of OAC, an important determinant of post-bleed outcomes, also varies by bleeding site, suggesting that site-specific factors may influence prognosis. Further research is needed to clarify the risk factors and outcomes associated with specific bleed sites.³⁶

Our analysis, which includes individual participant data collected prospectively from randomized controlled trials, represents the largest and highest-quality dataset of systematically adjudicated OAC-related extracranial bleeding among patients with AF analyzed through an individual patient meta-analysis. The availability of detailed baseline characteristics allowed comprehensive population characterization, adjustment analyses, identification of risk factors, and estimation of population attributable risk. Nonetheless, several limitations should be acknowledged. While participants enrolled in randomized controlled trials tend to differ from unselected patients treated in routine care, thereby limiting direct generalizability, our analysis includes over 70 000 patients across large international studies, providing a broad and clinically relevant population rather than a narrowly selected subgroup. Although the identified associations are likely to be clinically informative, we acknowledge that absolute bleeding rates and the estimated contribution of individual risk factors may differ in less selected real-world

populations. With respect to the regression models, some bleeding risk factors (eg, liver disease, alcohol consumption, and prior bleeding) were not included, and residual confounding may persist despite adjustment. Although bleeding outcomes were standardized and adjudicated, thereby enhancing the robustness of our results, the RE-LY trial did not capture CRNM bleeding events as per ISTH criteria, raising the possibility that some bleeds that would have met criteria for ISTH CRNM bleeding would have been missed; a sensitivity analysis excluding RE-LY showed similar findings as the main analysis. The high degree of missingness for some variables that were not collected in some trials could influence model estimates and corresponding aPAF calculations. We used the Ferguson (logistic regression) method for estimating aPAF, which does not incorporate follow-up time. However, this method allows estimation of the contribution of each risk factor individually and collectively, accounting for overlap between risk factors, which was the objective of our analysis.

Future research should further explore the prognostic implications of extracranial clinically relevant bleeding, including its association with mortality, recurrent bleeding, subsequent ischemic stroke or systemic embolism, and patient-reported outcomes (eg, quality of life) post-bleed to enhance evidence-informed decision-making.

Conclusion

Extracranial clinically relevant bleeding is common among OAC-treated patients with AF and may better capture the overall burden of bleeding complications experienced by patients compared with major bleeding. However, in our large cohort, analysis of extensive baseline characteristics did not entirely account for bleeding risk, suggesting that unmeasured or unknown factors contribute to bleeding risk. Future research should aim to characterize outcomes after clinically relevant bleeding, clarify its consequences for patients, and investigate novel risk factors (eg, biomarkers) to further improve risk stratification and guide clinical decision-making.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S9

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